Balancing Equations and Conservation of Mass

PhET Balancing Chemical Equations · PhET

Year 9	~30 minutes	Science · Chemistry

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This worksheet was drafted automatically from the site’s curriculum-mapping data and has not been checked by a human curriculum expert. Please review every task for accuracy, safety and suitability for your specific class before use.

Simulation: https://phet.colorado.edu/en/simulations/balancing-chemical-equations
Curriculum links: AC9S9U07 · AC9S8U07 · chemical reactions

Learning intentions

- I can balance simple chemical equations by adjusting coefficients.
- I can use molecule counts to explain why a chemical equation must be balanced.
- I can connect balanced equations to the law of conservation of mass.

Getting started

Open PhET Balancing Chemical Equations and start on the 'Introduction' screen. Each screen shows an unbalanced equation with molecule pictures above it and a balance scale below — the scale will only sit level when your equation is correctly balanced.

Tasks

1. Look at the first unbalanced equation. Count the atoms of each element on the left and right side. Are they equal? How can you tell from the balance scale?

2. Adjust the coefficients until the balance scale is level. Write down the balanced equation you ended up with.

3. Balance four different equations in the simulation and record the coefficients you used for each.

Equation	Coefficients used	Balanced? (Y/N)

4. Explain in your own words why a chemical equation has to be 'balanced' — what real scientific law does this represent?

5. Balance three new equations, working carefully rather than guessing. Which one took you the most tries, and what made it tricky?

6. For one balanced equation from your table, count the total number of atoms of each element on the reactant side and the product side. Show that the totals match.

Working:

7. A student says 'balancing an equation means making the number of molecules equal on both sides.' Explain why this statement is incorrect, using conservation of mass and atom counts to correct it.

8. Give one everyday example (e.g. burning, rusting, cooking) where a chemical reaction occurs, and explain what stays conserved even though the substances look completely different afterwards.

Extension (optional)

The equation for photosynthesis is unbalanced as written: $CO_2 + H_2O \rightarrow C_6H_{12}O_6 + O_2$. Without using the simulation, try to work out sensible coefficients that would balance the carbon, hydrogen and oxygen atoms, and explain your reasoning.